Linear regression: Abalone growth and pH example

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Note: Make sure a code and data folder are separated, this keeps things organized.

Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization, use if data is in a folder
library(readxl) # reading .xlsx files
library(ggeffects) # generating model predictions
```

```r
library(flextable) # creating tables

# abalone data from Hamilton et al. 2022
abalone <- read_xlsx(here("data", "Abalone IMTA_growth and pH.xlsx"))
```

Abalone example

Data from Hamilton et al. 2022. "Integrated multi-trophic aquaculture mitigates the effects of ocean acidification: Seaweeds raise system pH and improve growth of juvenile abalone." https://doi.org/10.1016/j.aquaculture.2022.738571

a. Questions and hypotheses

Question: How does pH predict abalone growth (measured in change in shell surface area per day, mm^2 d^{-1})?

H_0: **pH does not affect abalone growth in change in shell surface area per day**

H_A: **pH does affect abalone growth in change in shell surface area per day**

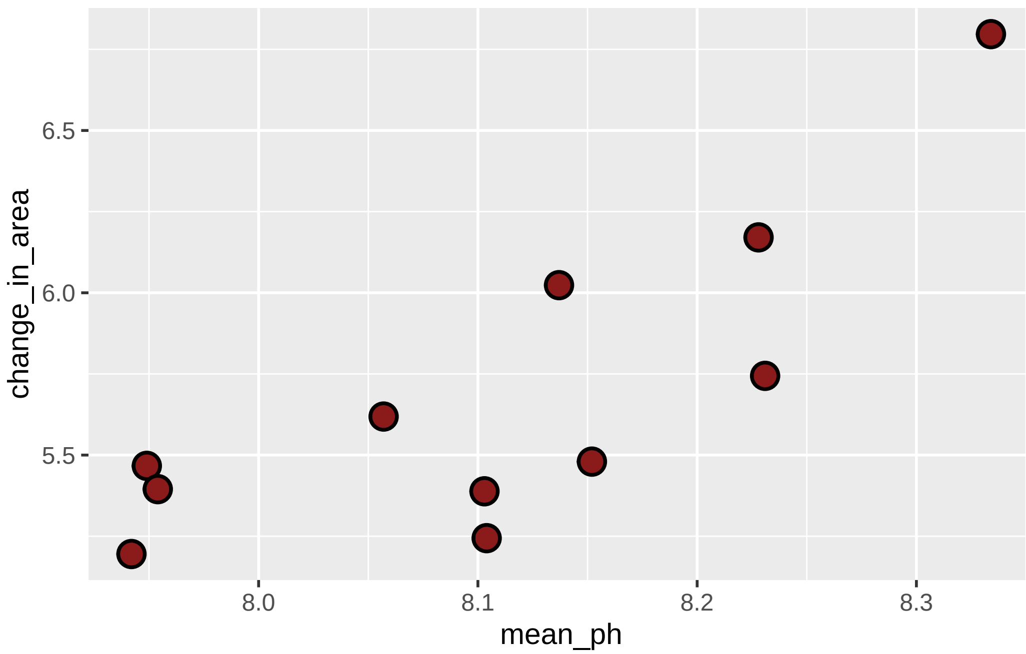
b. Cleaning

```r
# creating new object from abalone data
abalone_clean <- abalone |>
  # remove capitals, add underscores
  clean_names() |>
  # select the columns of interest
  # mean pH and total change in shell surface area
  select(mean_p_h, change_in_area_mm2_d_1_25) |>
  # rename columns to make it easier to understand
  rename(mean_ph = mean_p_h,
         change_in_area = change_in_area_mm2_d_1_25)
```

Don't forget to look at your data! Use `View(abalone_clean)` in the Console.

c. Exploratory data visualization

```
# create base layer
ggplot(data = abalone_clean, # use data from abalone_clean
       aes(x = mean_ph, # x-axis is the pH, y-axis is the change in area
           y = change_in_area)) +
  # first layer: create scatterplot using the observations
  geom_point(size = 4,
             stroke = 1,
             fill = "firebrick4",
             shape = 21)
```



Where does the missing value come from?

The missing value corresponds to the missing pH value. The pH sensor malfunctioned in one pair of tanks, leading to no pH reading here.

Don't forget to turn your warnings off either in the individual code chunk or in the YAML at the top of the file.

d. Abalone model

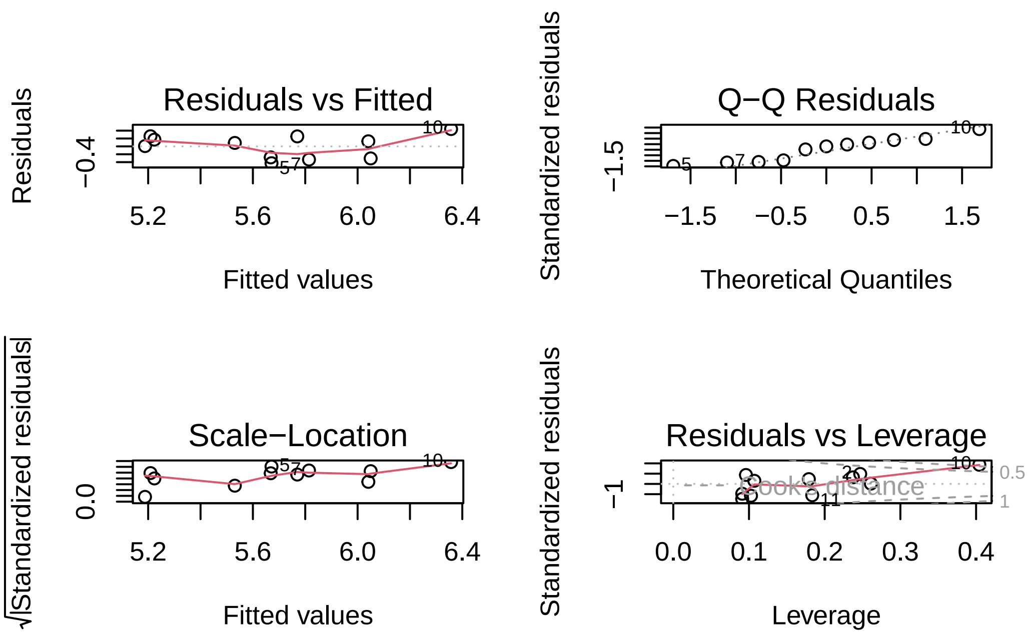
Model fitting

```
abalone_model <- lm(
  change_in_area ~ mean_ph, # formula: change in shell surface area as a function of pH
  data = abalone_clean      # data: use abalone_clean dataframe
)
```

Note: To view the model, run it in the console.

Diagnostics

```
par(mfrow = c(2, 2))    # displays plots in a 2x2 grid, ONLY way these show up correctly
plot(abalone_model)     # display diagnostic plots for abalone_model
```



Model summary

```r
# more information about the model
summary(abalone_model)
```

```

Call:
lm(formula = change_in_area ~ mean_ph, data = abalone_clean)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.42714 -0.29282  0.08769  0.21258  0.43965 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept) -18.5010     6.1601  -3.003  0.01488 * 
mean_ph       2.9827     0.7596   3.926  0.00348 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3063 on 9 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.6314,	Adjusted R-squared:  0.5905 
F-statistic: 15.42 on 1 and 9 DF,  p-value: 0.003477
```

Checks: Median near 0, 1st and 3rd quantiles are roughly equidistant from 0.

First column of numbers gives us the model estimates. Then standord errors, t values and p values for the estimates. We can see all p values are below 0.05, so there's confidence that neither the slope or y-int is 0.

To assess if the whole model is good, use the p-value from the f distribution (at the bottom). Ours is 0.003.

What is the slope estimate (with standard error?)

β_1 = 2.98 $\pm$ 0.76

What is the intercept estimate (with standard error?)

β_1 = -18.50 $\pm$ 6.16

Generating predictions

```
# creating a new object called abalone_preds
abalone_preds <- ggpredict(
  abalone_model,      # model object
  terms = "mean_ph"   # predictor (in quotation marks)
)

# display the predictions
abalone_preds
```

```
# Predicted values of change_in_area

mean_ph | Predicted |     95% CI
--------------------------------
   7.94 |      5.19 | 4.83, 5.54
   7.95 |      5.21 | 4.86, 5.55
   8.06 |      5.53 | 5.30, 5.76
   8.10 |      5.67 | 5.46, 5.88
   8.10 |      5.67 | 5.46, 5.88
   8.14 |      5.77 | 5.55, 5.98
   8.15 |      5.81 | 5.59, 6.04
   8.33 |      6.36 | 5.92, 6.80
```

Look at the column names using `colnames(abalone_preds)` in the Console.

What are the column names?

x, predicted, std.error, conf.low, conf.high, group (ignore this column for now)

Look at the actual object (by clicking on it in Environment).

Look at the class of the object using `class(abalone_preds)` in the Console.

What is this type of object?

ggeffects object and a data frame

View `abalone_preds` like a data frame. This means we can plot this like how we did during data visualization

Why generate model predictions?

This dataset does not include any actual observations of abalone growth at pH = 8.

However, the model can tell us a prediction for what abalone growth could be if pH = 8 based on the existing data.

Figure out what abalone growth the model would predict at pH = 8.

```
# finding the model prediction at a specific value
ggpredict(
  abalone_model,       # model object
  terms = "mean_ph[8]"    # predictor (in quotation marks) and predictor value in brackets
)
```

```
# Predicted values of change_in_area

mean_ph | Predicted |     95% CI
--------------------------------
      8 |      5.36 | 5.08, 5.64
```

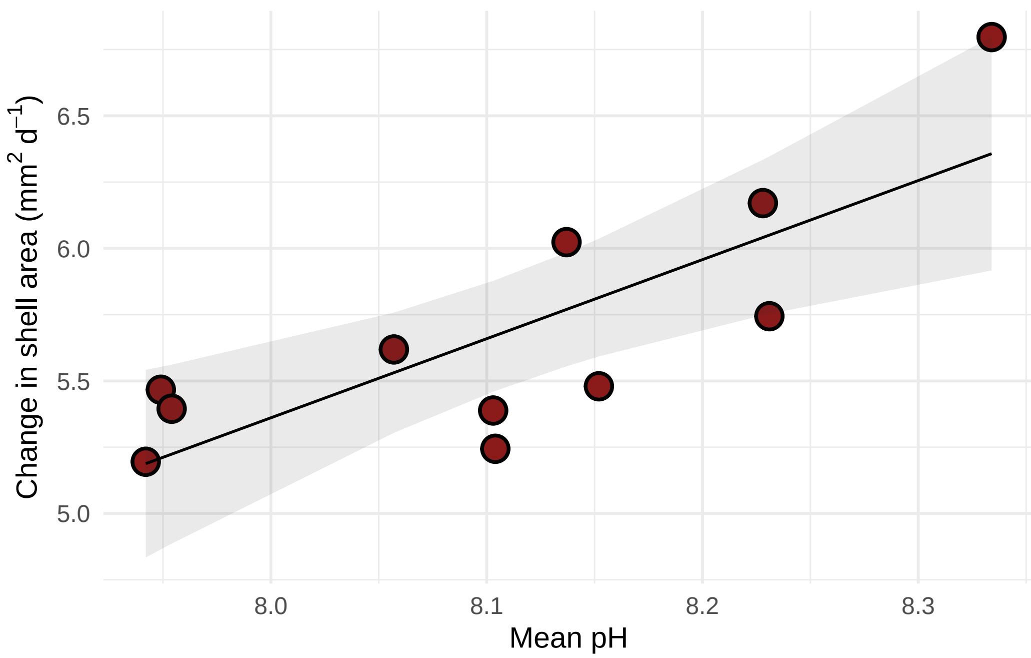
How would you interpret this?

At pH = 8, the model predicts abalone shell growth of 5.36 [5.08, 5.64] mm^2 d^{-1}.

Visualizing model predictions and data

```
# base layer: ggplot
# using clean data frame
ggplot(data = abalone_clean,
       aes(x = mean_ph,
           y = change_in_area)) +
  # first layer: points representing abalones
  geom_point(size = 4,
             stroke = 1,
             fill = "firebrick4",
             shape = 21) +
  # second layer: ribbon representing confidence interval
  # using predictions data frame
  geom_ribbon(data = abalone_preds,
              aes(x = x,
```

```
                 y = predicted,
                 ymin = conf.low,
                 ymax = conf.high),
             alpha = 0.1) +
  # third layer: line representing model predictions
  # using predictions data frame
  geom_line(data = abalone_preds,
            aes(x = x,
                y = predicted)) +
  # axis labels
  labs(x = "Mean pH",
       y = expression("Change in shell area ("*mm^{2}~d^-1*")")) +
  # theme
  theme_minimal()
```



Creating a table with model coefficients, 95% confidence intervals, and more

Note: this function is from `flextable`.

```
as_flextable(abalone_model)
```

	Estimate	Standard Error	t value	Pr(>|t|)	
(Intercept)	-18.501	6.160	-3.003	0.0149	*
mean_ph	2.983	0.760	3.926	0.0035	**

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 0.3063 on 9 degrees of freedom

Multiple R-squared: 0.6314, Adjusted R-squared: 0.5905

F-statistic: 15.42 on 9 and 1 DF, p-value: 0.0035

END OF ABALONE EXAMPLE